

Number System Conversions

Binary, Octal, Decimal, and Hexadecimal

Introduction

This assignment is focused to explain number system conversions. We will explore the four primary number systems – **Binary**, **Octal**, **Decimal**, and **Hexadecimal** – and learn how to convert numbers between them. Understanding these conversions is crucial in computer science and digital electronics, as different systems are used for different purposes.

Number Systems Overview

Decimal (Base-10)

The number system we use daily. It has ten digits (0-9).

Binary (Base-2)

The language of computers. It has two digits (0 and 1).

Octal (Base-8)

Uses eight digits (0-7). Often used as a shorthand for binary.

Hexadecimal (Base-16)

Uses sixteen digits (0-9 and A-F). Commonly used to represent memory addresses

Conversion Techniques

These are the 12 conversion processes, along with examples .

1. **Decimal to Binary**
2. **Decimal to Octal**
3. **Decimal to Hexadecimal**
4. **Binary to Decimal**
5. **Binary to Octal**
6. **Binary to Hexadecimal**
7. **Octal to Decimal**
8. **Octal to Binary**
9. **Octal to Hexadecimal**
10. **Hexadecimal to Decimal**
11. **Hexadecimal to Binary**

12. Hexadecimal to Octal

Decimal Conversions

1. Decimal to Binary

Method: Repeated division by 2. Record the remainders (0 or 1) at each step. The binary number is formed by reading the remainders from bottom to top.

Example: Convert 25 (decimal) to binary.

$$25 / 2 = 12, \text{ Remainder} = 1$$

$$12 / 2 = 6, \text{ Remainder} = 0$$

$$6 / 2 = 3, \text{ Remainder} = 0$$

$$3 / 2 = 1, \text{ Remainder} = 1$$

$$1 / 2 = 0, \text{ Remainder} = 1$$

Therefore, 25 (decimal) = 11001 (binary)

2. Decimal to Octal

Method: Repeated division by 8. Record the remainders (0-7) at each step. The octal number is formed by reading the remainders from bottom to top.

Example: Convert 150 (decimal) to octal.

$$150 / 8 = 18, \text{ Remainder} = 6$$

$$18 / 8 = 2, \text{ Remainder} = 2$$

$$2 / 8 = 0, \text{ Remainder} = 2$$

Therefore, 150 (decimal) = 226 (octal)

3. Decimal to Hexadecimal

Method: Repeated division by 16. Record the remainders (0-15). If the remainder is 10-15, represent it with the letters A-F (A=10, B=11, C=12, D=13, E=14, F=15). The hexadecimal number is formed by reading the remainders from bottom to top.

Example: Convert 450 (decimal) to hexadecimal.

$$450 / 16 = 28, \text{ Remainder} = 2$$

$$28 / 16 = 1, \text{ Remainder} = 12 \text{ (C)}$$

$1 / 16 = 0$, Remainder = 1

Therefore, 450 (decimal) = 1C2 (hexadecimal)

Binary Conversions

4. Binary to Decimal

Method: Multiply each digit by 2 raised to the power of its position (starting from 0 on the right) and sum the results.

Example: Convert 10110 (binary) to decimal.

$$(1 * 2^4) + (0 * 2^3) + (1 * 2^2) + (1 * 2^1) + (0 * 2^0) = 16 + 0 + 4 + 2 + 0 = 22$$

Therefore, 10110 (binary) = 22 (decimal)

5. Binary to Octal

Method: Group the binary digits into groups of three (from right to left). Add leading zeros if necessary. Convert each group of three into its octal equivalent.

Example: Convert 101101 (binary) to octal.

Grouped: 101 101

Octal equivalent: 5 5

Therefore, 101101 (binary) = 55 (octal)

6. Binary to Hexadecimal

Method: Group the binary digits into groups of four (from right to left). Add leading zeros if necessary. Convert each group of four into its hexadecimal equivalent.

Example: Convert 1101011 (binary) to hexadecimal.

Grouped: 0110 1011

Hexadecimal equivalent: 6 B

Therefore, 1101011 (binary) = 6B (hexadecimal)

Octal Conversions

7. Octal to Decimal

Method: Multiply each digit by 8 raised to the power of its position (starting from 0 on the right) and sum the results.

Example: Convert 237 (octal) to decimal.

$$(2 * 8^2) + (3 * 8^1) + (7 * 8^0) = 128 + 24 + 7 = 159$$

Therefore, 237 (octal) = 159 (decimal)

8. Octal to Binary

Method: Convert each octal digit to its 3-bit binary equivalent.

Example: Convert 72 (octal) to binary.

$$7 = 111$$

$$2 = 010$$

Therefore, 72 (octal) = 111010 (binary)

9. Octal to Hexadecimal

Method: First, convert the octal number to binary. Then, convert the binary number to hexadecimal.

Example: Convert 64 (octal) to hexadecimal.

$$64 \text{ (octal)} = 110100 \text{ (binary)}$$

$$110100 \text{ (binary)} = 34 \text{ (hexadecimal)}$$

Therefore, 64 (octal) = 34 (hexadecimal)

Hexadecimal Conversions

10. Hexadecimal to Decimal

Method: Multiply each digit by 16 raised to the power of its position (starting from 0 on the right) and sum the results. Remember that A=10, B=11, C=12, D=13, E=14, F=15.

Example: Convert 2A (hexadecimal) to decimal.

$$(2 * 16^1) + (10 * 16^0) = 32 + 10 = 42$$

Therefore, 2A (hexadecimal) = 42 (decimal)

11. Hexadecimal to Binary

Method: Convert each hexadecimal digit to its 4-bit binary equivalent.

Example: Convert A5 (hexadecimal) to binary.

A = 1010

5 = 0101

Therefore, A5 (hexadecimal) = 10100101 (binary)

12. Hexadecimal to Octal

Method: First, convert the hexadecimal number to binary. Then, convert the binary number to octal.

Example: Convert 1A (hexadecimal) to octal.

1A (hexadecimal) = 00011010 (binary)

00011010 (binary) = 032 (octal)

Therefore, 1A (hexadecimal) = 32 (octal)

Conclusion

These are the complete 12 conversions from all the number system to one and another